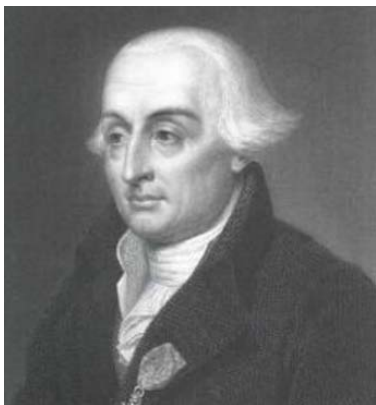




[Joseph-Louis Lagrange](#) (1736~1813)

法国著名数学家、物理学家。

拉格朗日科学研究所涉及的领域极其广泛。他在数学上最突出的贡献是使数学分析与几何与力学脱离开来，使数学的独立性更为清楚，从此数学不再仅仅是其他学科的工具。

6.2 拉格朗日插值

主讲： 施明辉

厦门大学

6.2 拉格朗日插值

- 6.2.1 线性插值与抛物插值
- 6.2.2 拉格朗日插值
- 6.2.3 插值余项与误差估计

6.2.1 线性插值与抛物插值

1. 线性插值

问题：已知两点 (x_0, y_0) 及 (x_1, y_1) ，求插值多项式。

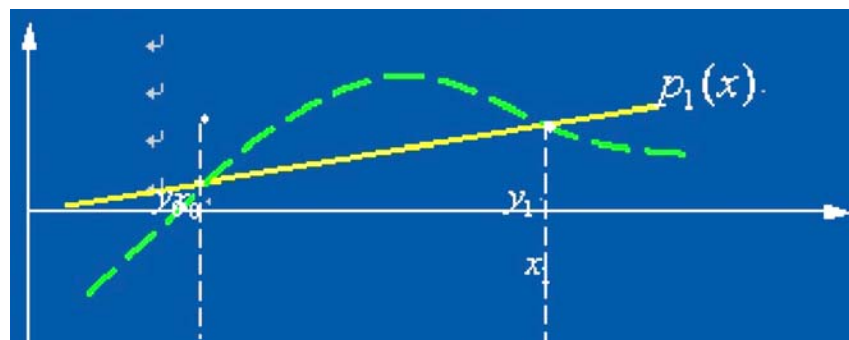
过两点的插值多项式是一条直线。

两点式直线方程：

$$L_1(x) = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1$$

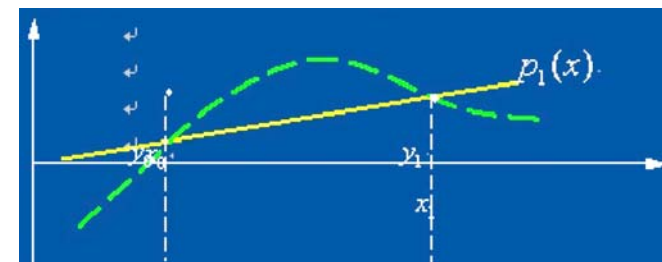
显然， $L_1(x_0)=y_0$ ， $L_1(x_1)=y_1$ 。

$L_1(x)$ 满足插值条件，是线性插值函数。



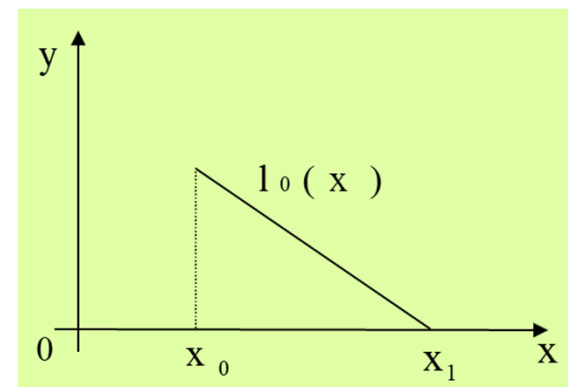
$$L_1(x) = \underbrace{\frac{x - x_1}{x_0 - x_1}}_{l_0(x)} y_0 + \underbrace{\frac{x - x_0}{x_1 - x_0}}_{l_1(x)} y_1$$

插值基函数 → $l_0(x)$



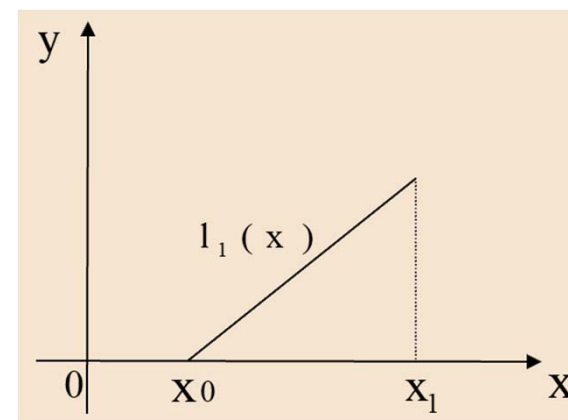
$$l_0(x) = \frac{x - x_1}{x_0 - x_1}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0}$$



$$L_1(x) = y_0 l_0(x) + y_1 l_1(x) \quad (6.2.1)$$

$$L_1(x) = \sum_{i=0}^1 l_i(x) y_i$$

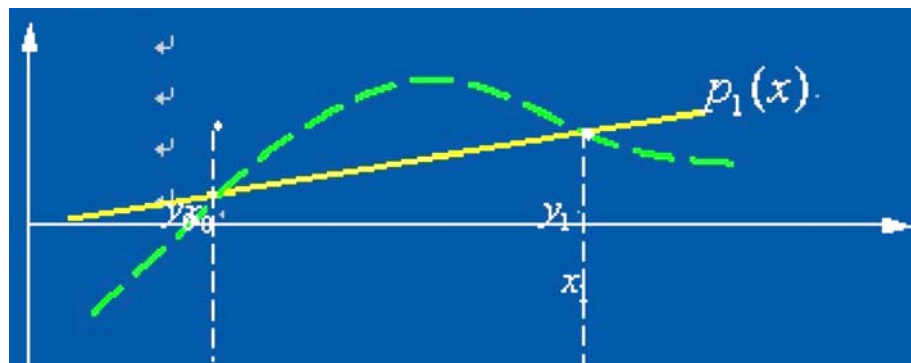


$$L_1(x) = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1$$

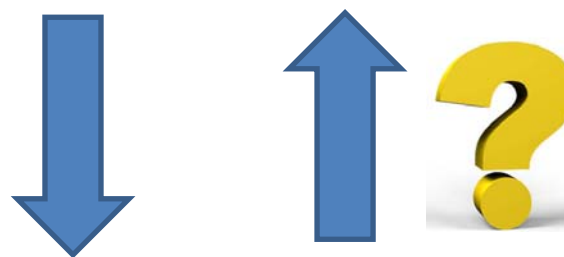
例 6.2.1: 已知 $\sqrt{100} = 10, \sqrt{121} = 11$ 求 $y = \sqrt{115}$

解: 设 $x_0 = 100, y_0 = 10, x_1 = 121, y_1 = 11$

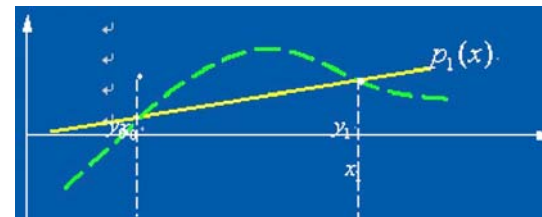
$$y = \sqrt{115} \approx \frac{115 - 121}{100 - 121} \times 10 + \frac{115 - 100}{121 - 100} \times 11 = 10.71428$$



$$L_1(x) = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1$$



$$L_1(x) = y_0 l_0(x) + y_1 l_1(x)$$



$$L_1(x) = y_0 l_0(x) + y_1 l_1(x)$$

这里 $l_0(x)$ 和 $l_1(x)$ 分别看作是满足条件:

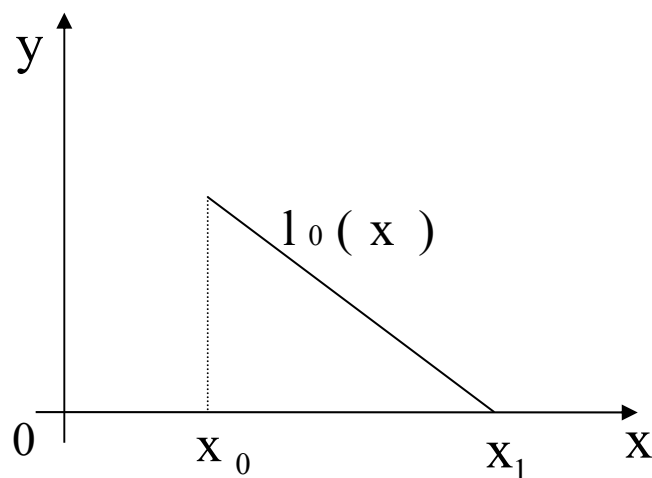
$$\begin{cases} l_0(x_0) = 1, l_0(x_1) = 0 \\ l_1(x_0) = 0, l_1(x_1) = 1 \end{cases}$$

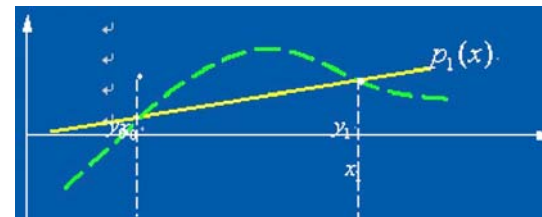
$$l_0(x) = c(x - x_1) \quad l_0(x_0) = 1$$

$$l_0(x) = \frac{(x - x_1)}{(x_0 - x_1)} \quad c = \frac{1}{(x_0 - x_1)}$$

显然, $L_1(x_0) = y_0, L_1(x_1) = y_1$ 。

$L_1(x)$ 满足插值条件, 是线性插值函数。





$$L_1(x) = y_0 l_0(x) + y_1 l_1(x)$$

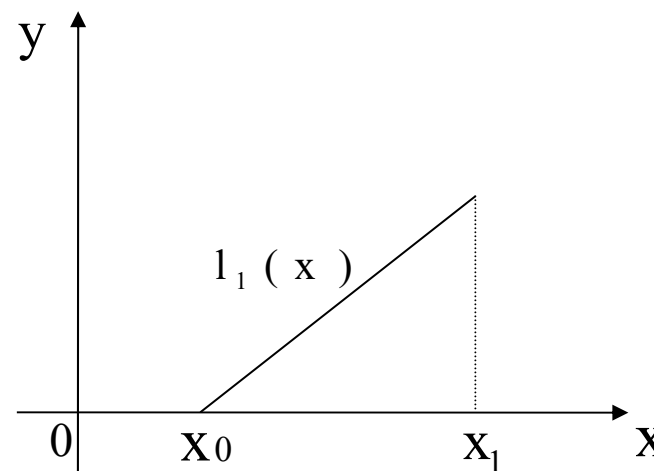
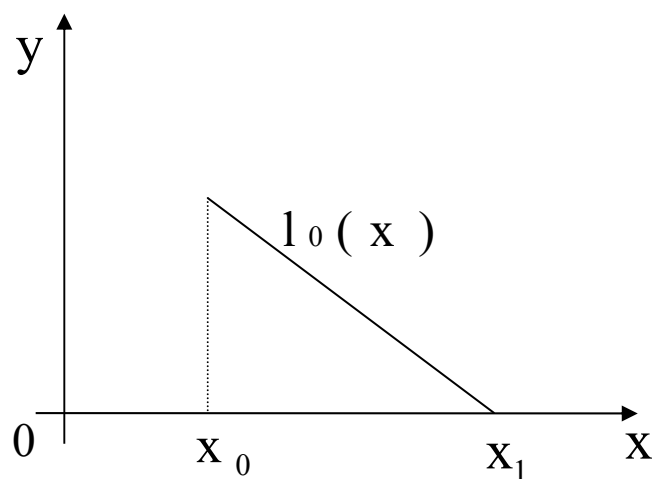
这里 $l_0(x)$ 和 $l_1(x)$ 分别看作是满足条件:

$$\begin{cases} l_0(x_0) = 1, l_0(x_1) = 0 \\ l_1(x_0) = 0, l_1(x_1) = 1 \end{cases}$$

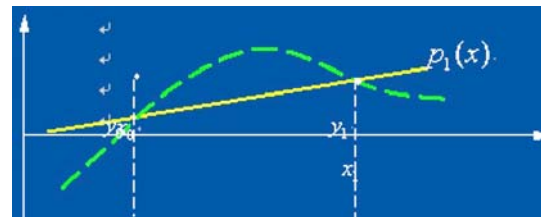
$$l_1(x) = \frac{(x - x_0)}{(x_1 - x_0)} \quad c = \frac{1}{(x_1 - x_0)}$$

显然, $L_1(x_0) = y_0, L_1(x_1) = y_1$ 。

$L_1(x)$ 满足插值条件, 是线性插值函数。



$$L_1(x) = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1$$



↑

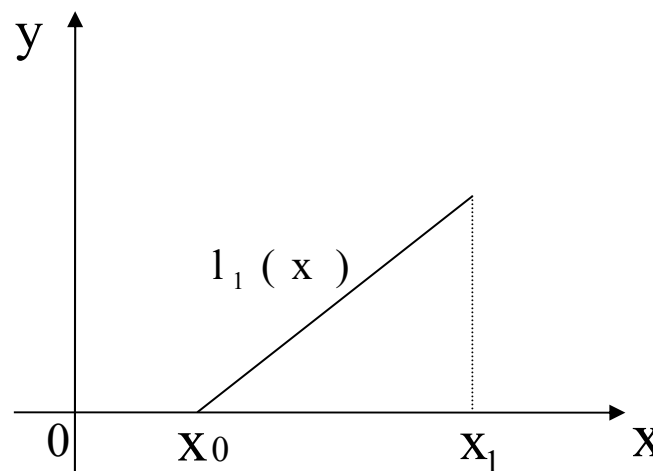
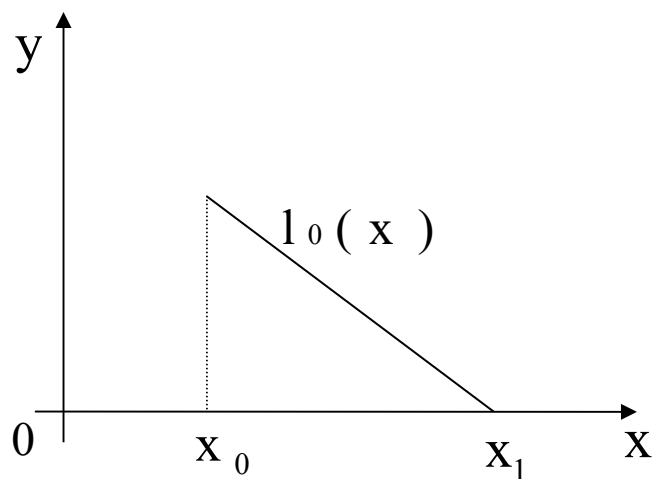
$$L_1(x) = y_0 l_0(x) + y_1 l_1(x) \quad l_0(x) = \frac{(x - x_1)}{(x_0 - x_1)} \quad l_1(x) = \frac{(x - x_0)}{(x_1 - x_0)}$$

这里 $l_0(x)$ 和 $l_1(x)$ 分别看作是满足条件:

$$\begin{cases} l_0(x_0) = 1, l_0(x_1) = 0 \\ l_1(x_0) = 0, l_1(x_1) = 1 \end{cases}$$

显然, $L_1(x_0) = y_0, L_1(x_1) = y_1$ 。

$L_1(x)$ 满足插值条件, 是线性插值函数。

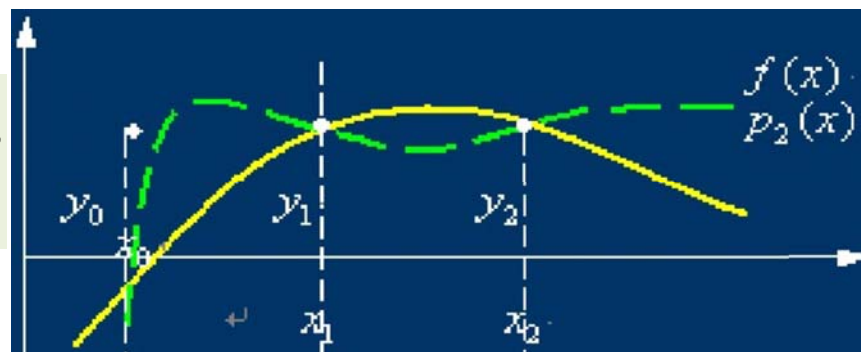


6.2.1 线性插值与抛物插值

1. 抛物插值

问题：已知三点 (x_0, y_0) 、 (x_1, y_1) 及 (x_2, y_2) ，求插值多项式。

过三点的插值多项式是一条抛物线，也称为**抛物插值函数**。

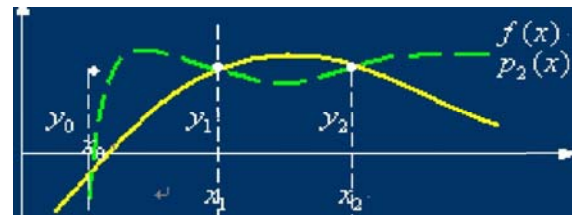


$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$

↑ 推广 ?

$$L_1(x) = y_0 l_0(x) + y_1 l_1(x)$$

$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$



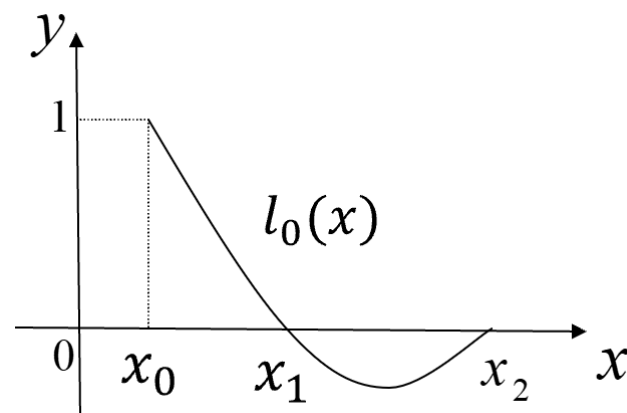
$$\begin{cases} l_0(x_0) = 1, & l_0(x_1) = 0, & l_0(x_2) = 0 \\ l_1(x_0) = 0, & l_1(x_1) = 1, & l_1(x_2) = 0 \\ l_2(x_0) = 0, & l_2(x_1) = 0, & l_2(x_2) = 1 \end{cases} \quad (6.2.4)$$

$$L_2(x_0) = y_0 \quad L_2(x_1) = y_1 \quad L_2(x_2) = y_2$$

$$l_0(x) = c(x - x_1)(x - x_2)$$

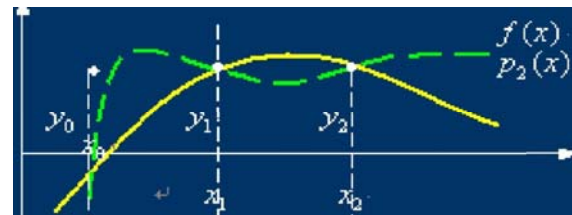
$$l_0(x_0) = 1$$

$$c = \frac{1}{(x_0 - x_1)(x_0 - x_2)}$$



$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$



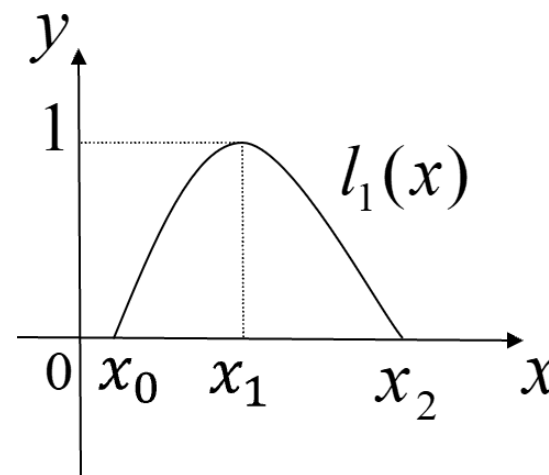
$$\begin{cases} l_0(x_0) = 1, & l_0(x_1) = 0, & l_0(x_2) = 0 \\ l_1(x_0) = 0, & l_1(x_1) = 1, & l_1(x_2) = 0 \\ l_2(x_0) = 0, & l_2(x_1) = 0, & l_2(x_2) = 1 \end{cases} \quad (6.2.4)$$

$$L_2(x_0) = y_0 \quad L_2(x_1) = y_1 \quad L_2(x_2) = y_2$$

$$l_1(x) = c(x - x_0)(x - x_2)$$

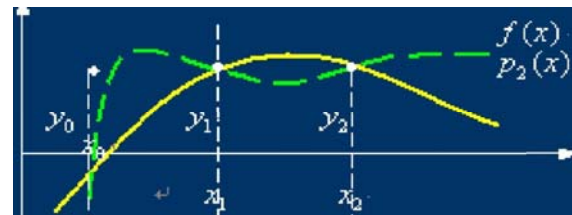
$$l_1(x_1) = 1$$

$$c = \frac{1}{(x_1 - x_0)(x_1 - x_2)}$$



$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$



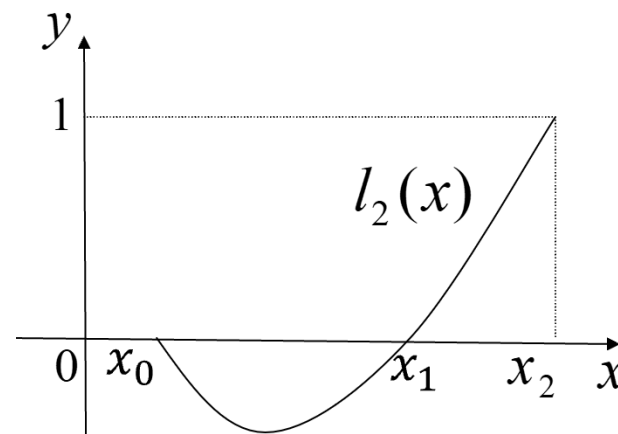
$$\begin{cases} l_0(x_0) = 1, & l_0(x_1) = 0, & l_0(x_2) = 0 \\ l_1(x_0) = 0, & l_1(x_1) = 1, & l_1(x_2) = 0 \\ l_2(x_0) = 0, & l_2(x_1) = 0, & l_2(x_2) = 1 \end{cases} \quad (6.2.4)$$

$$L_2(x_0) = y_0 \quad L_2(x_1) = y_1 \quad L_2(x_2) = y_2$$

$$l_2(x) = c (x - x_0) (x - x_1)$$

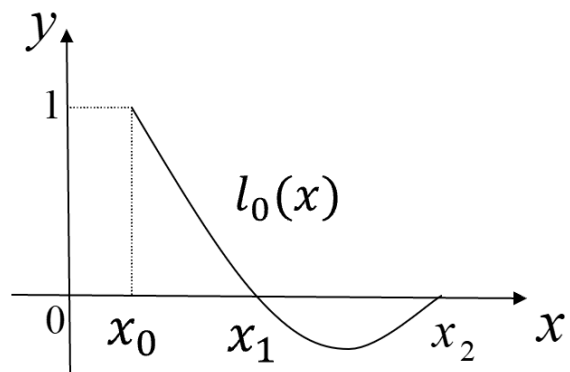
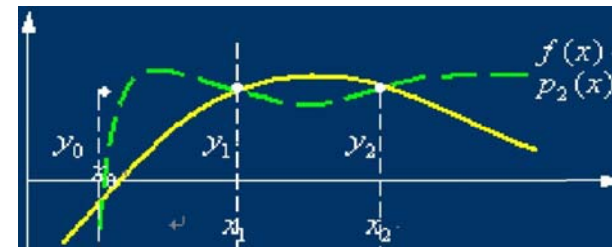
$$l_2(x_2) = 1$$

$$c = \frac{1}{(x_2 - x_0)(x_2 - x_1)}$$

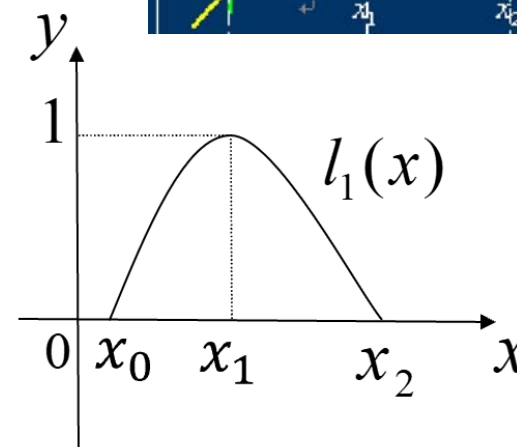


$$l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$

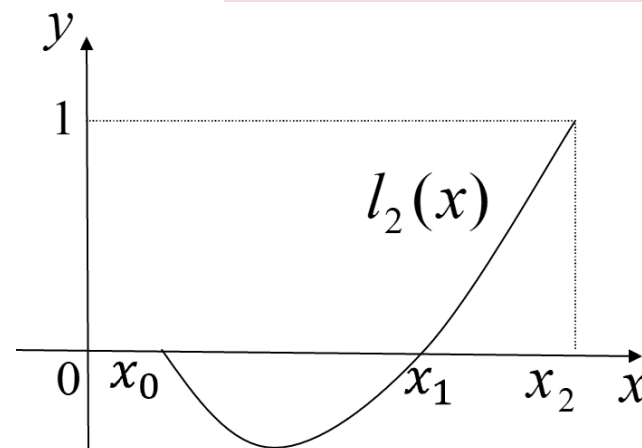


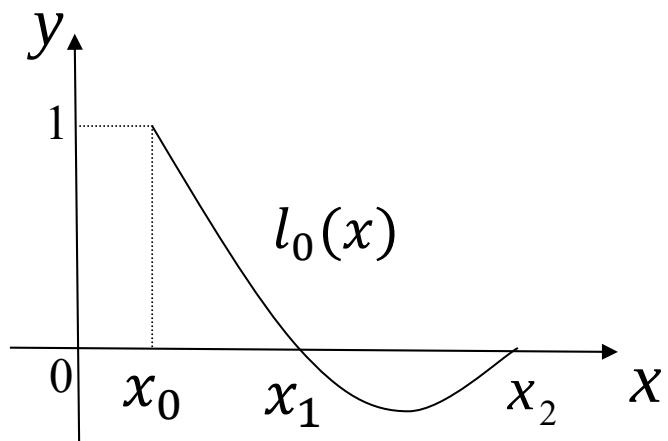
$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$



$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

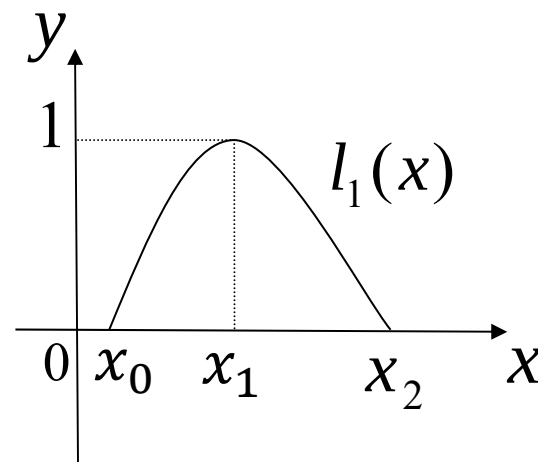
$$l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$



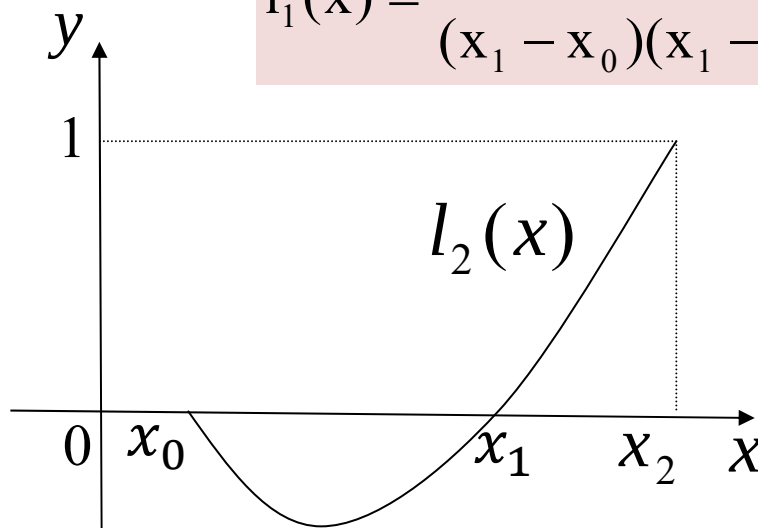


$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$



$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

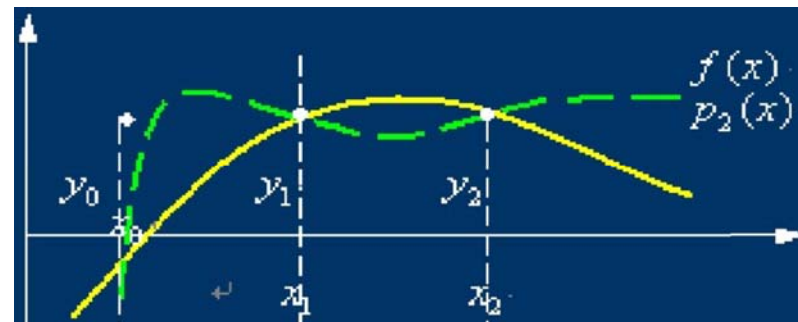


$$L_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$

$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$



$$L_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

(6.2.5)

$$L_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

例6.2.2 已知 $\sqrt{100}=10$, $\sqrt{121}=11$, $\sqrt{144}=12$ 求 $\sqrt{115}$

解: 设 $x_0=100$, $y_0=10$, $x_1=121$, $y_1=11$, $x_2=144$, $y_2=12$

将 $x=115$ 代入(6.2.5)式, 得

$$y = \sqrt{115} \approx$$

$$\frac{(115-121)(115-144)}{(100-121)(100-144)} \times 10 + \frac{(115-100)(115-144)}{(121-100)(121-144)} \times 11 + \frac{(115-100)(115-121)}{(144-100)(144-121)} \times 12$$

$$= 10.7228$$



$$L_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

例6.2.2 已知 $\sqrt{100}=10$, $\sqrt{121}=11$, $\sqrt{144}=12$ 求 $\sqrt{115}$

解：设 $x_0=100$, $y_0=10$, $x_1=121$, $y_1=11$, $x_2=144$, $y_2=12$

令 $x=115$ 代入 (6.2.5) 式, 得

$$y = \sqrt{115} \approx$$

$$\frac{(115-121)(115-144)}{(100-121)(100-144)} \times 10 + \frac{(115-100)(115-144)}{(121-100)(121-144)} \times 11 + \frac{(115-100)(115-121)}{(144-100)(144-121)} \times 12$$

$$= 10.7228$$



6.2.2 拉格朗日插值

x	x_0	x_1	x_2	$\dots\dots$	x_n
$y=f(x)$	y_0	y_1	y_2	$\dots\dots$	y_n

已知 $n+1$ 个点 (x_i, y_i) ($i=0, 1, 2, \dots, n$),

构造 n 次插值多项式 $L_n(x)$

使得 $L_n(x_i) = y_i$ ($i=0, 1, 2, \dots, n$)

用插值基函数的方法, 可设

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (6.2.6)$$

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (6.2.6)$$

其中, $l_i(x)$ 称为关于 $x_0, x_1, \dots, x_n$ 的插值基函数, 它满足条件:

$$l_i(x_j) = \begin{cases} 1, j = i \\ 0, j \neq i \end{cases} \quad i, j = 0, 1, \dots, n \quad (6.2.7)$$

$$L_n(x_k) = \sum_{i=0}^n l_i(x_k) y_i = y_k$$

线性插值
($n=1$)

$$\begin{cases} l_0(x_0) = 1, l_0(x_1) = 0 \\ l_1(x_0) = 0, l_1(x_1) = 1 \end{cases}$$

抛物插值
($n=2$)

$$\begin{cases} l_0(x_0) = 1, & l_0(x_1) = 0, & l_0(x_2) = 0 \\ l_1(x_0) = 0, & l_1(x_1) = 1, & l_1(x_2) = 0 \\ l_2(x_0) = 0, & l_2(x_1) = 0, & l_2(x_2) = 1 \end{cases}$$

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (6.2.6)$$

其中, $l_i(x)$ 称为关于 $x_0, x_1, \dots, x_n$ 的插值基函数, 它满足条件:

$$l_i(x_j) = \begin{cases} 1, j = i \\ 0, j \neq i \end{cases} \quad i, j = 0, 1, \dots, n \quad (6.2.7)$$

这表明, 除 x_i 外的所有节点都是 $l_i(x)$ 的零点,

$$l_i(x) = c \prod_{\substack{j=0 \\ j \neq i}}^n (x - x_j)$$

再由条件 $l_i(x_i) = 1$ 确定其系数 c

$$l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} = \frac{(x - x_0) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_0) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)}$$

6.2.2 拉格朗日插值

x	x_0	x_1	x_2	$\dots\dots$	x_n
$y=f(x)$	y_0	y_1	y_2	$\dots\dots$	y_n

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (6.2.6)$$

$$\begin{aligned} l_i(x) &= \frac{(x-x_0)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)} \\ &= \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x-x_j}{x_i-x_j} \end{aligned} \quad (6.2.8)$$

$$L_n(x) = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x-x_j}{x_i-x_j} \right) y_i \quad (6.2.9)$$

例 6.2.3 已知 $x=1,2,3,4,5$ 对应的函数值为 $f(x)=1,4,7,8,6$,
试构造 4 次拉格朗日插值多项式, 并求 $f(1.5)$ 的近似值。

$$\begin{aligned} L_4(x) = & \frac{(x-2)(x-3)(x-4)(x-5)}{(1-2)(1-3)(1-4)(1-5)} \times 1 + \frac{(x-1)(x-3)(x-4)(x-5)}{(2-1)(2-3)(2-4)(2-5)} \times 4 + \\ & \frac{(x-1)(x-2)(x-4)(x-5)}{(3-1)(3-2)(3-4)(3-5)} \times 7 + \frac{(x-1)(x-2)(x-3)(x-5)}{(4-1)(4-2)(4-3)(4-5)} \times 8 + \\ & \frac{(x-1)(x-2)(x-3)(x-4)}{(5-1)(5-2)(5-3)(5-4)} \times 6 = \cdots = \frac{1}{24}x^4 - \frac{3}{4}x^3 + \frac{83}{24}x^2 - \frac{11}{4}x + 1 \end{aligned}$$

$$f(1.5) \approx L_4(1.5) = \frac{299}{128} = 2.3359375$$

输入 $n, x, (x_i, y_i) \ (i=0,1,2,\dots,n)$

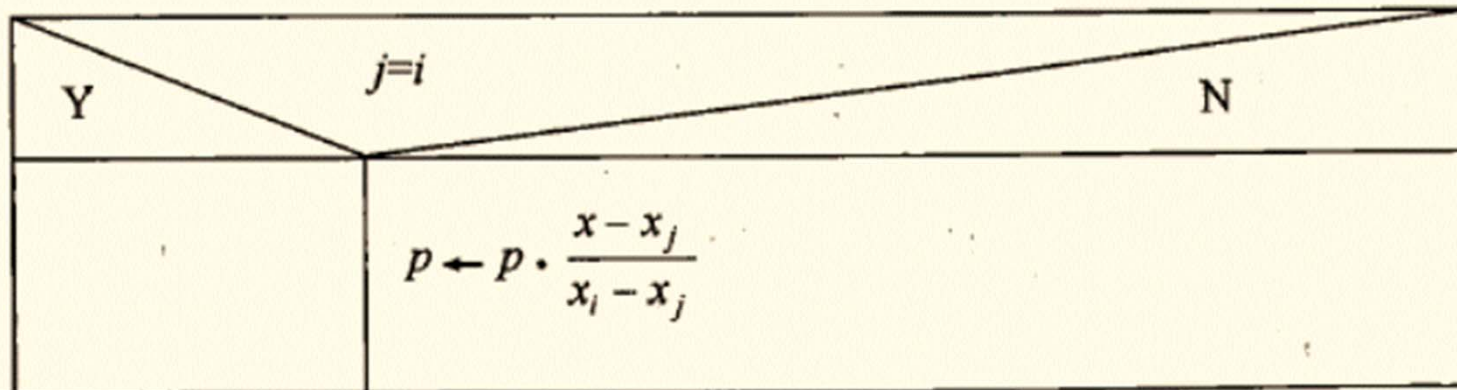
$y \leftarrow 0$

对于 $i=0,1,2,\dots,n$ 执行

$$L_n(x) = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \right) y_i \quad (6.2.9)$$

$p \leftarrow 1$

对于 $j=0,1,2,\dots,n$ 执行



$y = y + p \cdot y_i$

输出 x, y



6.2.3 插值余项与误差估计

$$R_n(x) = f(x) - L_n(x)$$

定理6.2.1

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \prod_{j=0}^n (x - x_j)$$

6.2.3 插值余项与误差估计

定理6.2.1 设 $f(x)$ 在 $[a, b]$ 上 $(n+1)$ 阶导数连续,
记 $f(x) \in C^{n+1}[a, b]$ 且 $f(x)$ 在互异节点

$$a \leq x_0 < x_1 < \dots < x_n \leq b$$

的函数值为 $y_0, y_1, \dots, y_n$ 若满足插值条件 $L_n(x_i) = y_i$
($i=0, 1, 2, \dots, n$)的插值多项式为 $L_n(x)$,

则对 $\forall x \in [a, b]$ 有:

$$\omega_{n+1}(x) = \prod_{j=0}^n (x - x_j)$$

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \prod_{j=0}^n (x - x_j) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \omega_{n+1}(x)$$

其中 $a < \xi < b$ (6.2.10)

6.2.3 插值余项与误差估计

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \prod_{j=0}^n (x - x_j) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \omega_{n+1}(x)$$

其中 $a < \xi < b$ (6.2.10)

$$\left| f^{(n+1)}(\xi) \right| \leq \left| \max_{a \leq x \leq b} f^{(n+1)}(x) \right| \leq M_{n+1}$$

$$\left| R_n(x) \right| \leq \frac{M_{n+1}}{(n+1)!} \left| \prod_{j=0}^n (x - x_j) \right| \quad (6.2.12)$$

$$|R_n(x)| \leq \frac{M_{n+1}}{(n+1)!} \left| \prod_{j=0}^n (x - x_j) \right| \quad (6.2.12)$$

当 $n=1$ 时，线性插值式 (6.2.1) 的误差估计式为



参见例 6.2.4

$$|R_1(x)| \leq \frac{M_2}{2!} |(x - x_0)(x - x_1)| \leq \frac{M_2}{8} (x_1 - x_0)^2 \quad (6.2.13)$$

当 $n=2$ 时，抛物插值 (6.2.5) 式的误差估计式为：

$$|R_2(x)| \leq \frac{M_3}{3!} |(x - x_0)(x - x_1)(x - x_2)| \quad (6.2.14)$$

例 6.2.4 设 $f(x) \in C^2[a, b]$, 已知插值节点 $x_0 = a, x_1 = b$,
证明: $f(x)$ 在 $[a, b]$ 内的线性插值函数 $L_1(x)$ 的误差界为

$$\max_{a \leq x \leq b} |f(x) - L_1(x)| \leq \frac{(b-a)^2}{8} \max_{a \leq x \leq b} |f''(x)|$$

 参见例 6.2.4

$$|R_1(x)| \leq \frac{M_2}{2!} |(x - x_0)(x - x_1)| \leq \frac{M_2}{8} (x_1 - x_0)^2 \quad (6.2.13)$$

例 6.2.4 设 $f(x) \in C^2[a, b]$, 已知插值节点 $x_0 = a, x_1 = b$, 证明: $f(x)$ 在 $[a, b]$ 内的线性插值函数 $L_1(x)$ 的误差界为

$$\max_{a \leq x \leq b} |f(x) - L_1(x)| \leq \frac{(b-a)^2}{8} \max_{a \leq x \leq b} |f''(x)|$$

证明 由插值余项公式, 可得 $R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \prod_{j=0}^n (x - x_j)$

$$f(x) - L_1(x) = \frac{f''(\xi)}{2!} (x-a)(x-b), \quad a < \xi < b.$$

$$|(x-a)(x-b)| \text{ 在 } x = \frac{a+b}{2} \text{ 处取得最大值为 } \frac{(b-a)^2}{4}$$

$$\begin{aligned} \max_{a \leq x \leq b} |f(x) - L_1(x)| &\leq \frac{1}{2} \max_{a \leq x \leq b} |(x-a)(x-b)| \max_{a \leq x \leq b} |f''(x)| \\ &= \frac{(b-a)^2}{8} \max_{a \leq x \leq b} |f''(x)| \end{aligned}$$

例6. 2. 5: $\sqrt{100} = 10$, $\sqrt{121} = 11$, $\sqrt{144} = 12$

对例6. 2. 1线性插值和例6. 2. 2抛物插值进行误差估计。

$$|R_1(x)| \leq \frac{M_2}{2!} |(x - x_0)(x - x_1)| \leq \frac{M_2}{8} (x_1 - x_0)^2 \quad (6. 2. 13)$$

$$|R_2(x)| \leq \frac{M_3}{3!} |(x - x_0)(x - x_1)(x - x_2)| \quad (6. 2. 14)$$

解：由题意知被插函数 为 $y = f(x) = \sqrt{x}$

$$f'(x) = \frac{1}{2} x^{-\frac{1}{2}} \quad f''(x) = -\frac{1}{4} x^{-\frac{3}{2}} \quad f'''(x) = \frac{3}{8} x^{-\frac{5}{2}}$$

$$x \in [100, 144]$$

$$M_2 = \max_{100 \leq x \leq 121} \left| -\frac{1}{4} x^{-\frac{3}{2}} \right| \leq \frac{1}{4000} \quad M_3 = \max_{100 \leq x \leq 144} \left| \frac{3}{8} x^{-\frac{5}{2}} \right| \leq \frac{3}{8} \times \frac{1}{100000}$$

于是

$$|R_1(x)| \leq \frac{M_2}{2!} |(x - x_0)(x - x_1)| \leq \frac{M_2}{8} (x_1 - x_0)^2 \quad (6.2.13)$$

$$|R_1(x)| \leq \left| \frac{1}{2} \times \frac{1}{4000} \times (115 - 100)(115 - 121) \right| = 0.01125 < 0.05$$

这说明例6.2.1中的近似值10.71428具有三位有效数字。

$$|R_2(x)| \leq \frac{M_3}{3!} |(x - x_0)(x - x_1)(x - x_2)| \quad (6.2.14)$$

$$\begin{aligned} |R_2(x)| &\leq \frac{1}{6} \times \frac{3}{800000} \times |(115 - 100)(115 - 121)(115 - 144)| \\ &= \frac{45 \times 29}{800000} = 0.00163125 \leq 0.005 \end{aligned}$$

这说明例6.2.2中的近似值10.7228具有四位有效数字。

例 6.2.6 设 $f(x) = x^4$ ，试利用拉格朗日插值余项定理写出以 $-1, 0, 1, 2$ 为插值节点的三次插值多项式。

解 设由已知插值节点确定的三次插值多项式为 $L_3(x)$ ，

$$f(x) - L_3(x) = \frac{f^{(4)}(\xi)}{4!} (x - x_0)(x - x_1)(x - x_2)(x - x_3) \quad (\xi \in (-1, 2))$$

由于 $f(x) = x^4$ ，则 $f^{(4)}(x) = 4!$ ，

$$\text{故 } f(x) - L_3(x) = (x+1)(x-0)(x-1)(x-2)$$

$$L_3(x) = x^4 - x(x+1)(x-1)(x-2) = 2x^3 + x^2 - 2x$$

例 6.2.7 设 x_j 为互异节点 ($j = 0, 1, \dots, n$), $l_j(x)$ 为 n 次插值基函数, 证明:

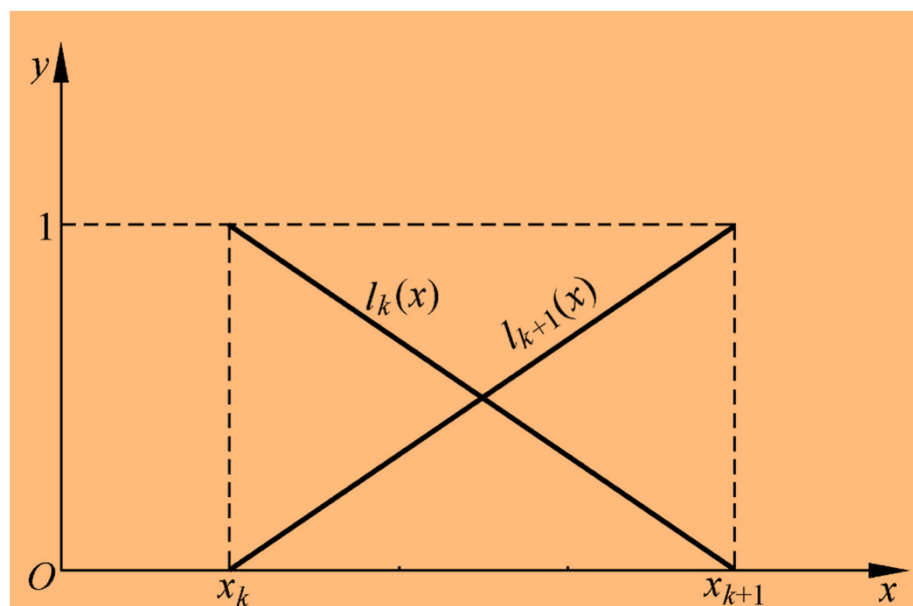
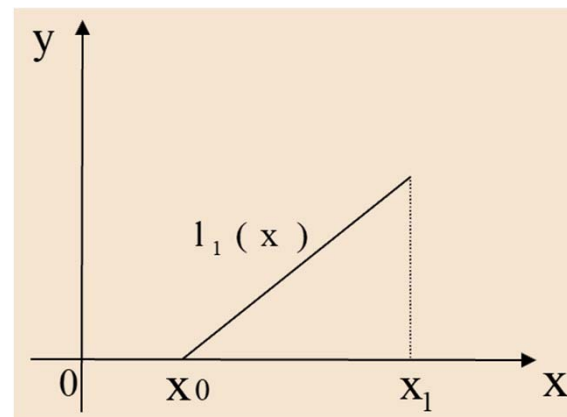
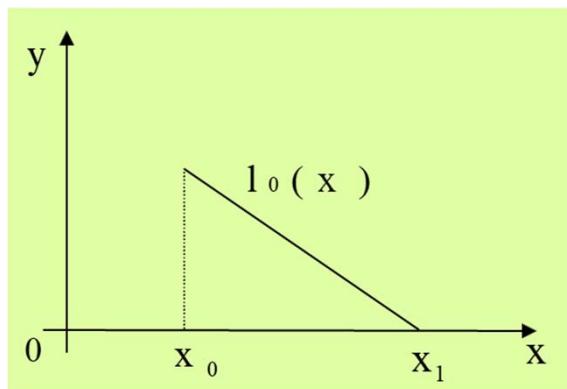
$$(1) \sum_{j=0}^n l_j(x) = 1$$

$$(2) \sum_{j=0}^n l_j(0) x_j^k = \begin{cases} 1 & k = 0 \\ 0 & k = 1, 2, \dots, n \\ (-1)^n x_0 x_1 \cdots x_n & k = n+1 \end{cases}$$

证明 (1) 对拉格朗日插值多项式, 令 $f(x) = 1$, 则得

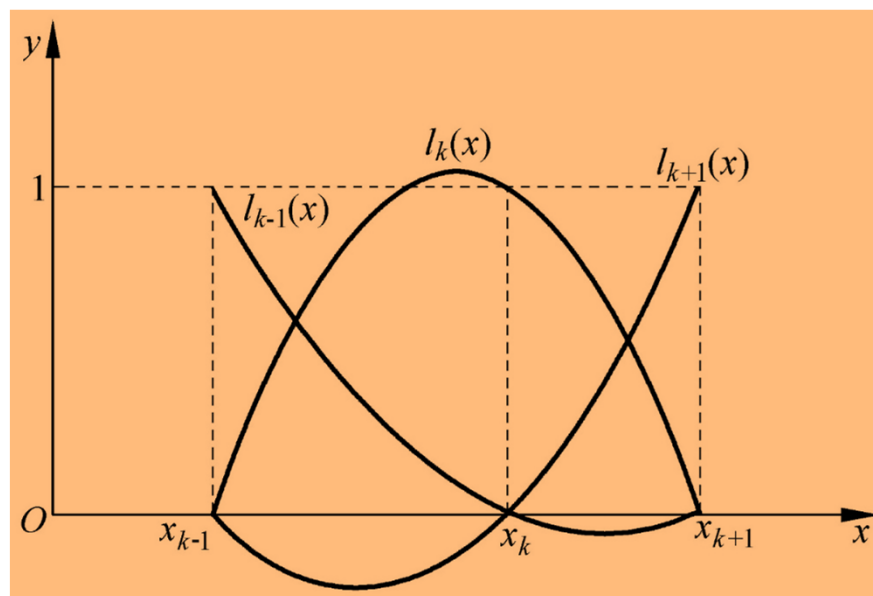
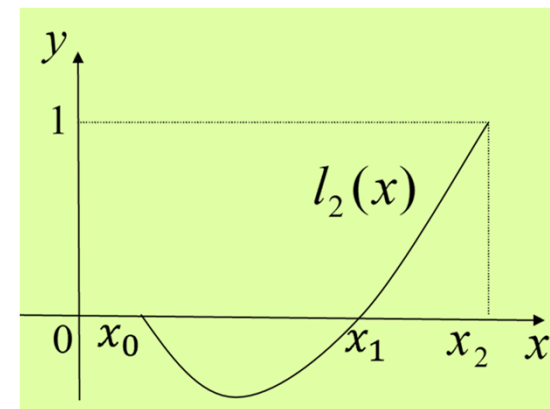
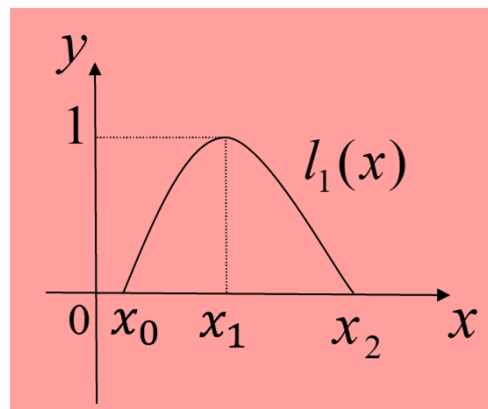
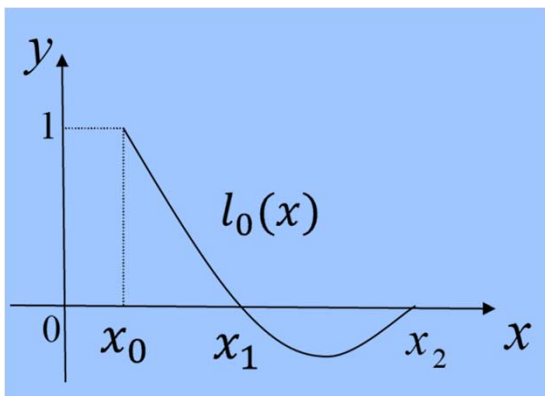
$$1 = \sum_{j=0}^n l_j(x) \times 1 + R_n(x)$$

因为 $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x) = 0$, 故有 $\sum_{j=0}^n l_j(x) = 1$ 。



$$\sum_{i=0}^n l_i(x) \equiv 1$$

线性插值基函数 $l_0(x)$, $l_1(x)$



$$\sum_{i=0}^n l_i(x) \equiv 1$$

抛物插值基函数 $l_0(x)$, $l_1(x)$, $l_2(x)$

$$(2) \quad \sum_{j=0}^n l_j(0)x_j^k = \begin{cases} 1 & k=0 \\ 0 & k=1,2,\cdots,n \\ (-1)^n x_0 x_1 \cdots x_n & k=n+1 \end{cases}$$

(2) 仍由 $f(x) = \sum_{j=0}^n l_j(x)f(x_j) + \frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}(x)$, 令 $f(x) = x^k$, 则

当 $k=0, f(x)=1$ 时, 有 $\sum_{j=0}^n l_j(0)=1$

当 $k=1,2,\cdots,n$ 时, $f^{(n+1)}(\xi)=0$, 故有 $\sum_{j=0}^n l_j(x)x_j^k = x^k$

当 $x=0$ 时, 得 $\sum_{j=0}^n l_j(0)x_j^k = 0$

当 $k=n+1$ 时, $f^{(n+1)}(\xi)=(n+1)!$, $x^{n+1} = \sum_{j=0}^n l_j(x)x_j^{n+1} + \omega_{n+1}(x)$

当 $x=0$ 时, 得 $\sum_{j=0}^n l_j(0)x_j^{n+1} = -(-x_0)(-x_1)\cdots(-x_n) = (-1)^n x_0 x_1 \cdots x_n$

$$\sum l_j(0) x_j^k = \begin{cases} 1 & k=0 \\ 0 & k=1, 2, \dots, n \\ (-1)^n x_0 x_1 \cdots x_n & k=n+1 \end{cases}$$

证：

$$\sum_{j=0}^n l_j(x) f(x_j) = f(x) - R_n(x) = f(x) - \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x)$$

当 $x=0$, $f(x)=x^k$ 时, 得

$$\sum_{j=0}^n l_j(0) x_j^k = 0^k - R_n(0) = - \frac{(x^k)^{(n+1)}|_{x=0}}{(n+1)!} \omega_{n+1}(0)$$

(1) 当 $k=0$ 时, 得

$$\sum_{j=0}^n l_j(0) x_j^k = \sum_{j=0}^n l_j(0) = 1$$

(2) 当 $k=1, 2, \dots, n$ 时, 得

$$\sum_{j=0}^n l_j(0) x_j^k = 0$$

(3) 当 $k=n+1$ 时, 得

$$\sum_{j=0}^n l_j(0) x_j^k = -\omega_{n+1}(0) = (-1)^{n+1} x_0 x_1 \cdots x_n = (-1)^n x_0 x_1 \cdots x_n$$

$$\sum l_j(0)x_j^k = \begin{cases} 1 & k = 0 \\ 0 & k = 1, 2, \dots, n \\ (-1)^n x_0 x_1 \cdots x_n & k = n+1 \end{cases}$$

证:

$$\sum_{j=0}^n l_j(x) f(x_j) = f(x) - R_n(x) = f(x) - \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x)$$

当 $x = 0, f(x) = x^k$ 时, 得

$$\sum_{j=0}^n l_j(0)x_j^k = 0^k - R_n(0) = -\frac{(x^k)^{(n+1)}|_{x=\xi}}{(n+1)!} \omega_{n+1}(0)$$

(1) 当 $k=0$ 时, 得

$$\sum_{j=0}^n l_j(0)x_j^k = \sum_{j=0}^n l_j(0) = 1$$

(2) 当 $k=1, 2, 3, \dots, n$ 时, 得

$$\sum_{j=0}^n l_j(0)x_j^k = 0$$

(3) 当 $k=n+1$ 时, 得

$$\sum_{j=0}^n l_j(0)x_j^k = -\omega_{n+1}(0) = (-1)^{n+2} x_0 x_1 \cdots x_n = (-1)^n x_0 x_1 \cdots x_n$$

拉格朗日插值 (小结)

x	x_0	x_1	x_2		x_n
$y=f(x)$	y_0	y_1	y_2		y_n

$$L_n(x) = \sum_{i=0}^n y_i l_i(x) \quad (6.2.6)$$

$$l_i(x) = \frac{(x - x_0) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_0) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)}$$

$$= \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad (6.2.8)$$

$$L_n(x) = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \right) y_i \quad (6.2.9)$$

拉格朗日插值 (小结)

$$L_n(x) = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \right) y_i \quad (6.2.9)$$

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \prod_{j=0}^n (x - x_j) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \omega_{n+1}(x)$$

$$|R_n(x)| \leq \frac{M_{n+1}}{(n+1)!} \left| \prod_{j=0}^n (x - x_j) \right| \quad (6.2.12)$$

- End